S7.1 Body

S7.1

S7.1					
Test	Stat	*p*	df	Dist	Reject
Box-Pierce Q	13.430	0.098	8	Chi2	No
H0: No serial correlation up to lag 8					
Ljung-Box Q	14.266	0.075	8	Chi2	No
H0: No serial correlation up to lag 8					

S7.2a Body

S7.2(A)

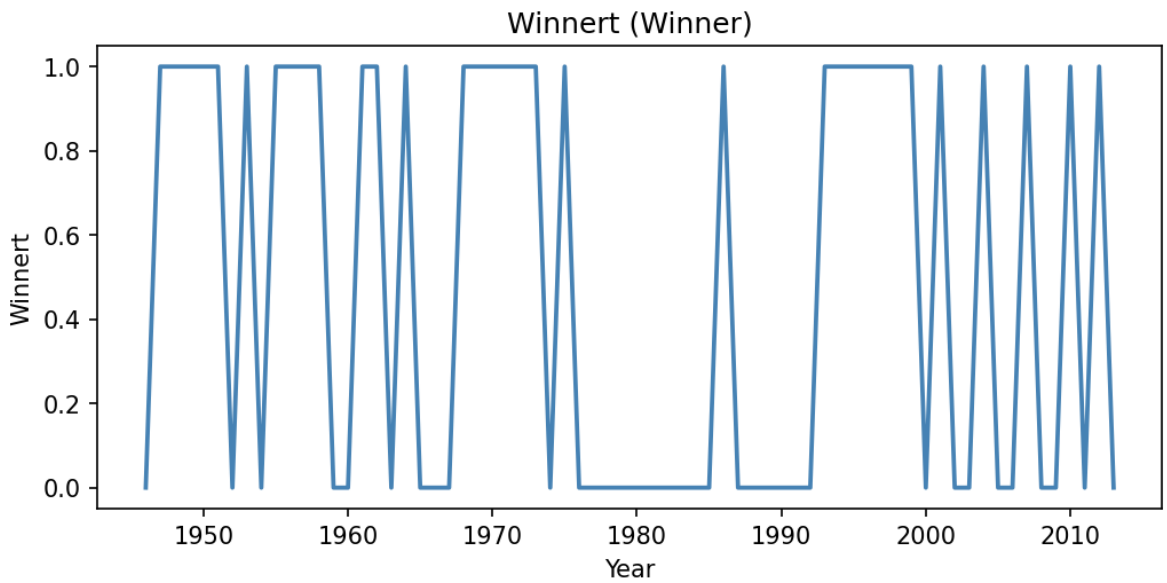

Figure 1: Time-series plot of Winnert (Winner)

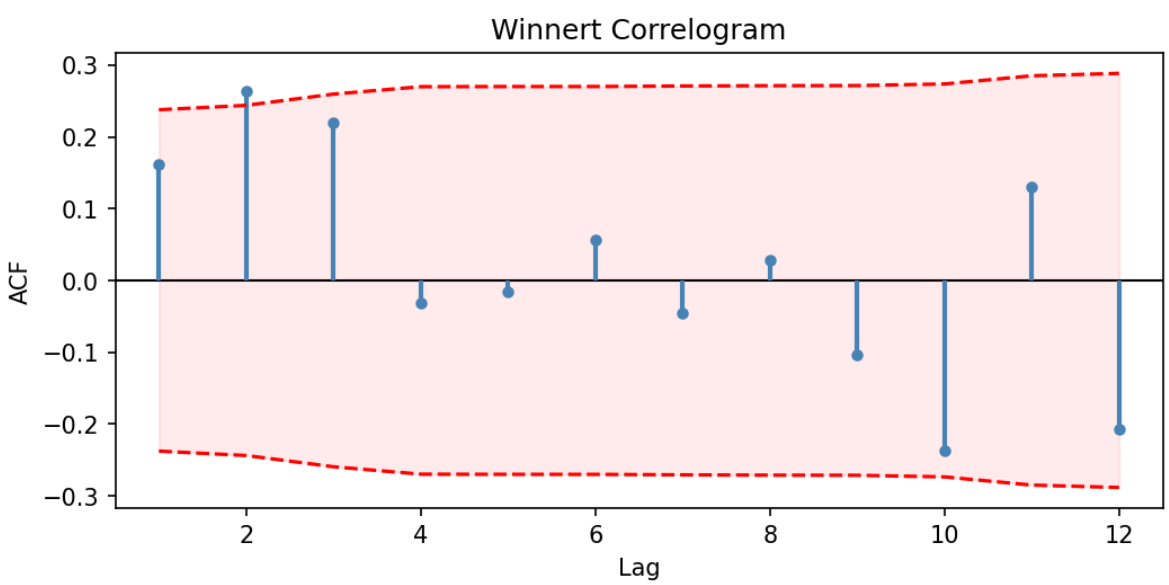

Figure 2: Sample correlogram of Winnert with Bartlett 95% bands (MA(0))

S7.2b Body

S7.2(B)

We test the null that the Winner series is white noise using the Ljung–Box test with $m = 4$. The test statistic is $Q_{LB} = 10.503$ with χ^2_4 reference distribution and p-value $= 0.033$. At the 5% level, we **reject** the null of white noise.

S7.2c Body

S7.2(C)

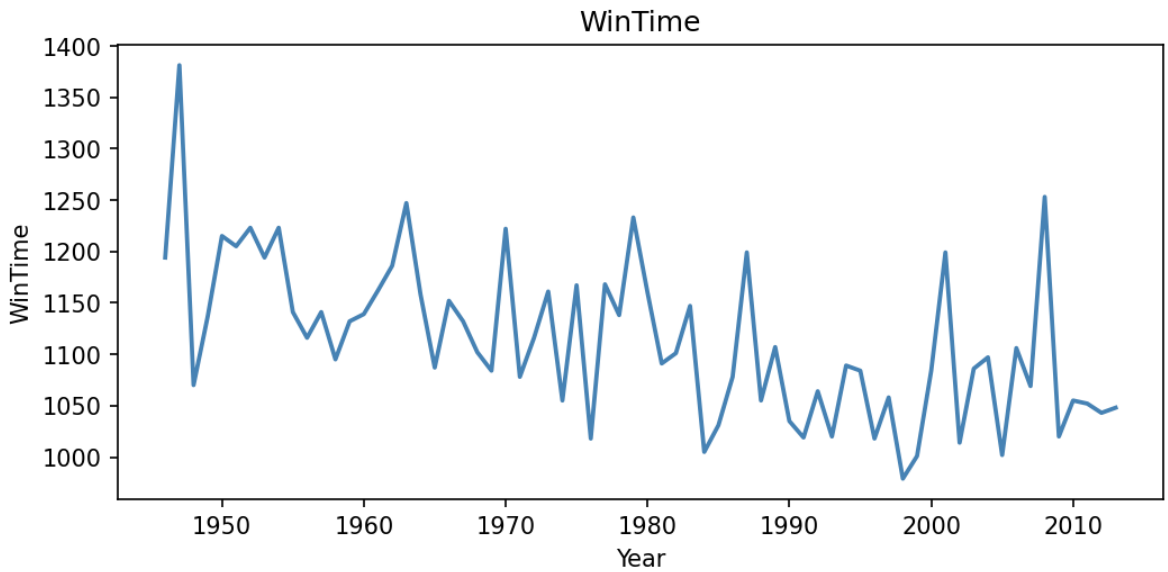

Figure 3: Time-series plot of WinTime

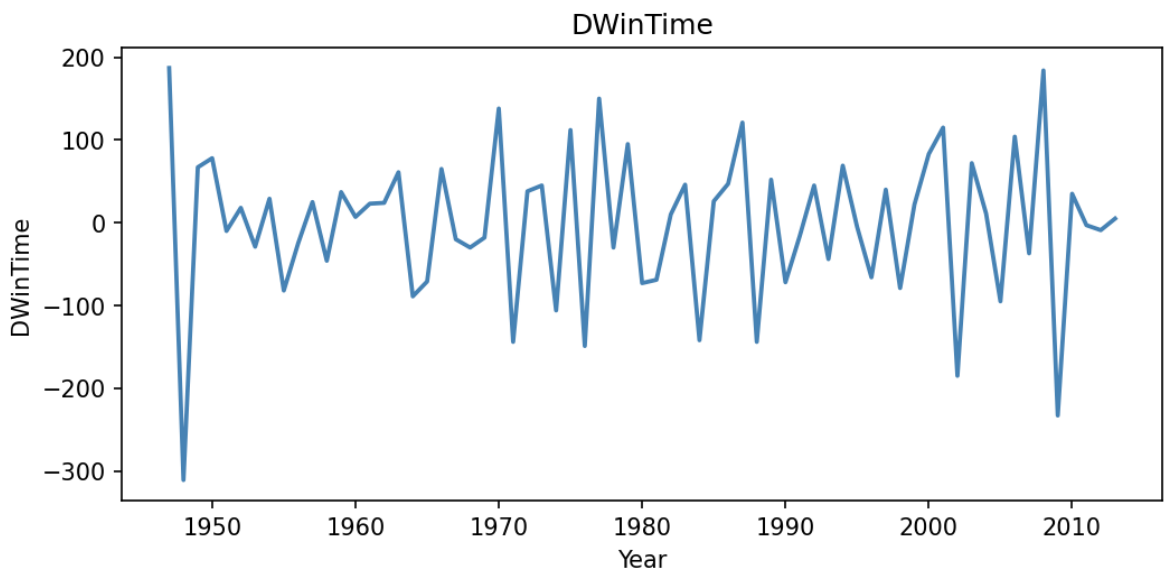

Figure 4: Time-series plot of DWinTime

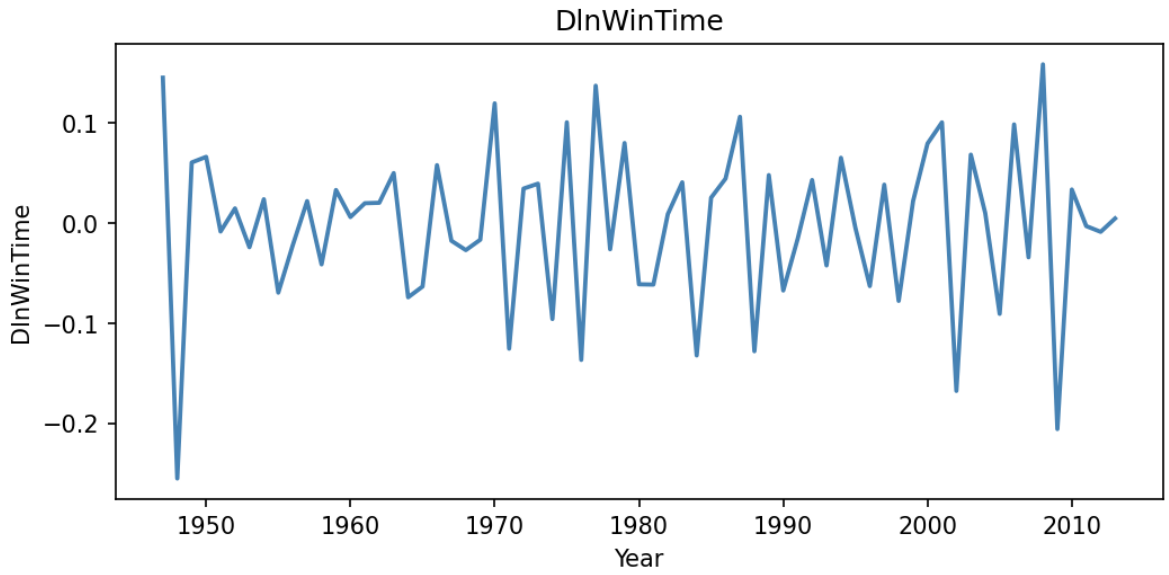

Figure 5: Time-series plot of DlnWinTime

WinTime shows moderate long-run variation with no obvious deterministic trend; year-to-year changes appear noisy. The first difference DWinTime fluctuates around zero with occasional spikes, suggesting short-run shocks. The log-difference DlnWinTime similarly centers near zero and compresses large swings, indicating proportionate changes are small and mean-reverting.

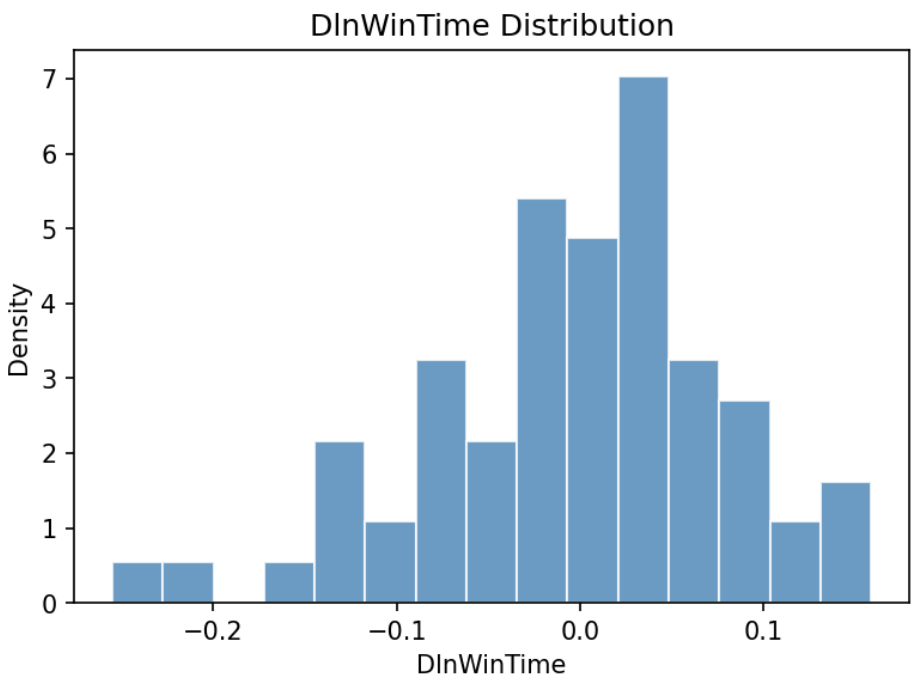

Figure 6: Distribution of DlnWinTime (histogram, density)

S7.2e Body

S7.2(E)

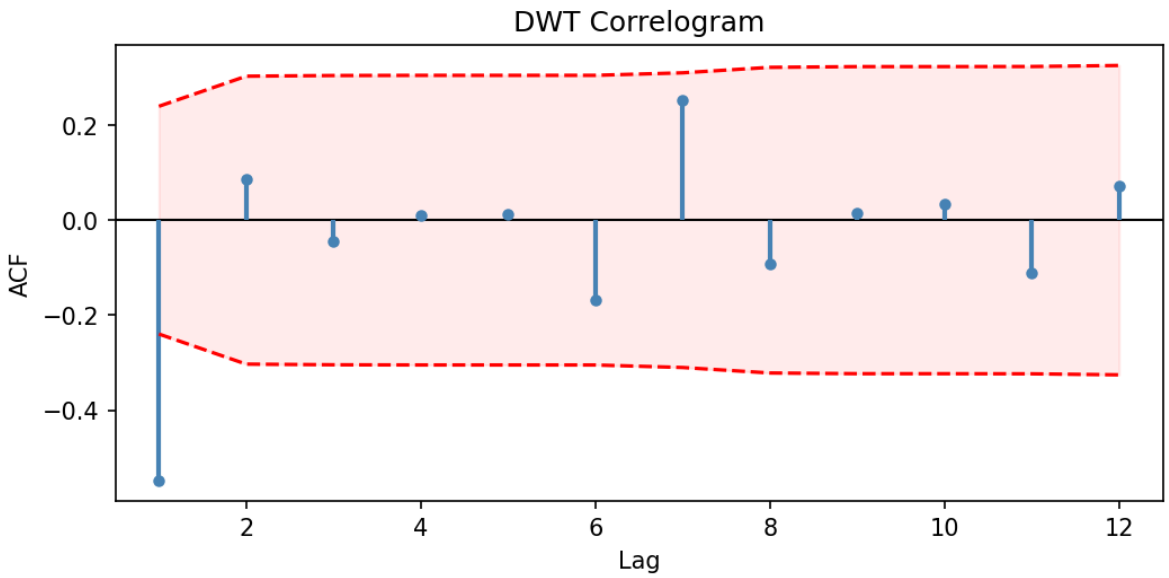

Figure 7: Sample correlogram of DWT (first difference of WinTime) with Bartlett 95% bands (MA(0))